1.9 Tangents to a Cubic

WORKSHEET

The question

STEP I 2009, Question 2

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/09-s1-q2

Hints

Try the question on your own first. If you get stuck, read the hints *one at a time*, and go back to the question after each one. Hints marked *Pause and try* are the key steps. There is space after each hint for your working.

Ask yourself The point $(-a, b)$ is on the curve. Why? And what does that have to do with sums of cubes?

STAGE 1 The tangent

Ask yourself Differentiate $y^3 = x^3 + a^3 + b^3$ implicitly. Remember that a and b are constants.

Pause and try Find the gradient at $(-a, b)$, write down the line through that point, and clear the fractions.

STAGE 2 Where the tangent meets the curve again

Ask yourself Write the tangent as $y = \ldots$ and substitute into $y^3 = x^3 + 9$.

Ask yourself Multiply out, collect everything on one side, and look for a common factor of the coefficients.

STAGE 3 Solving the cubic

Ask yourself You already know one root of the cubic, from the point of contact.

Pause and try A tangent touches the curve, so the point of contact is a *repeated* root. Divide out $(x+1)^2$ to find the third root, and the point where the tangent meets the curve again.

Ask yourself That point lies on $y^3 = x^3 + 9$. Clear the denominators. The leftover term is not a cube, but $9 = 1^3 + 2^3$.

What's really going on

The chord–tangent method

Take a cubic curve, such as $y^3 = x^3 + 9$ or the more standard form $y^2 = x^3 + Ax + B$. If you know two points on it with rational coordinates, the line through them meets

the curve in exactly one more point, and that point also has rational coordinates (by the argument in *Looking back*). If you know only one point, use the tangent there instead. Either way, old rational points produce new ones.

Remarkably, this "chord–tangent" construction can be made into a way of *adding* points on the curve, and the rational points then form a *group*, an algebraic structure you will study in your first year. Louis Mordell proved in 1922 that this group is generated by finitely many points: all the rational points, however many there are, can be built from a finite set using chords and tangents. His proof is a version of Fermat's descent from Section 1.5.

Elliptic curves today

Curves like these are called *elliptic curves* (not because they are ellipses, but because of a historical link with the arc length of an ellipse). They are everywhere in modern mathematics.

- **Fermat's Last Theorem.** In the 1980s Gerhard Frey noticed that a solution of $x^n + y^n = z^n$ would produce a very strange elliptic curve. Andrew Wiles proved in 1994 that such strange curves can't exist, and so proved Fermat's Last Theorem.
- **The Birch and Swinnerton-Dyer conjecture.** How many generators does the group of rational points need? A conjecture made in the 1960s, using some of the earliest computer experiments in mathematics at Cambridge, predicts the answer in terms of a function built from counting points modulo primes. It is one of the seven Millennium Prize Problems.
- **Cryptography.** On an elliptic curve over a finite number system, adding a point P to itself n times is quick, even when n has hundreds of digits (double P repeatedly, much as you would calculate a high power by repeated squaring). But given P and the result nP, finding n seems to be extremely hard: this is the *elliptic-curve discrete logarithm problem*. That asymmetry protects much of today's online communication.

Sums of like powers

The smallest cube that is a sum of three positive cubes is $6^3 = 3^3 + 4^3 + 5^3$; the tangent method found $20^3 = 7^3 + 14^3 + 17^3$, which is not a multiple of it, and repeating the method keeps producing new identities (although some involve negative numbers). How many kth powers does it take to make another kth power? Euler conjectured in 1769 that you need at least k kth powers to make another kth power. For fourth and fifth powers he was wrong: in 1966 Leon Lander and Thomas Parkin found

$$27^5 + 84^5 + 110^5 + 133^5 = 144^5$$

by computer search, and in 1988 Noam Elkies used elliptic curves to find counterexamples for fourth powers.

Try this next

1 **Rational points on a circle.** The line through $(-1, 0)$ with gradient t meets the circle $x^2 + y^2 = 1$ again. Show that the other point is $\left(\dfrac{1 - t^2}{1 + t^2}, \dfrac{2t}{1 + t^2}\right)$, and use this to produce infinitely many Pythagorean triples.

2 **A chord that misses.** The points $(1, 2)$ and $(2, 1)$ lie on $x^3 + y^3 = 9$. Show that the line through them meets the curve in no other point. Where has the "third point" gone? (Look up the *point at infinity*.)

3 **Repeat the trick.** Find the tangent to $y^3 = x^3 + 9$ at $\left(\frac{17}{7}, \frac{20}{7}\right)$ and the third point where it meets the curve. (A computer algebra system helps; notice how quickly the numbers grow.)

4 **Taxicab numbers.** $1729 = 1^3 + 12^3 = 9^3 + 10^3$. Find the third point where the line through $(1, 12)$ and $(9, 10)$ meets the curve $x^3 + y^3 = 1729$, and deduce a new identity of the form $p^3 = q^3 + r^3 + s^3$.

Further reading

- **Joseph Silverman & John Tate, *Rational Points on Elliptic Curves*** (Springer). Written for undergraduates; Chapter 1 needs nothing beyond A-level.
- **Avner Ash & Robert Gross, *Elliptic Tales*** (Princeton University Press). A popular account leading up to the Birch and Swinnerton-Dyer conjecture.
- **Simon Singh, *Fermat's Last Theorem*.** How elliptic curves came to solve a 350-year-old problem.

The full guided solution is in Section 1.9:
https://beyondstep.keithmcnulty.org/ch01/tangents-cubic.html